Customer Overview

Designed by : John Fafiolu

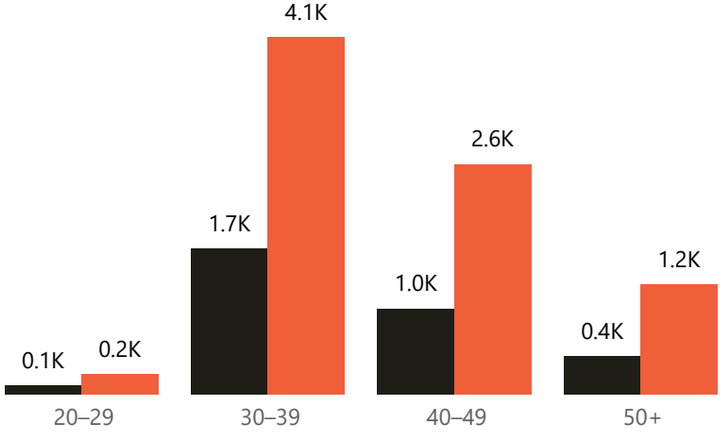
Total Customers	Churn Classification Rate	Avg. Act. card per Cust.
11,314	72.05%	0.90

Churn Labelled Customer	Not-Churned Customer	Average Age
8,152	3,162	40.46

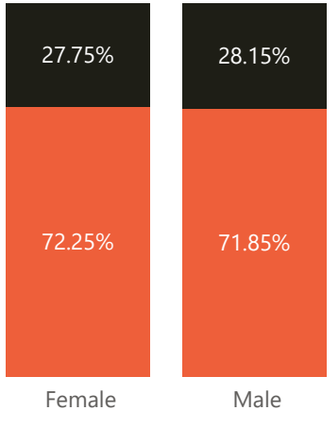
Customer Classification Distribution



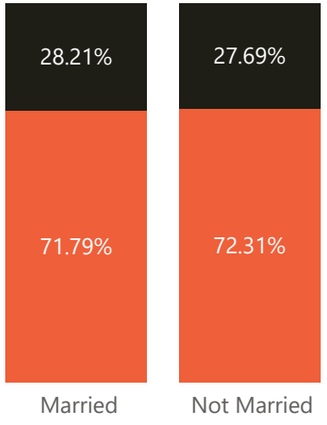
Customer Age Distribution by Classification.



Customer Classification by Gender



Average Transaction Count by Customer Classification

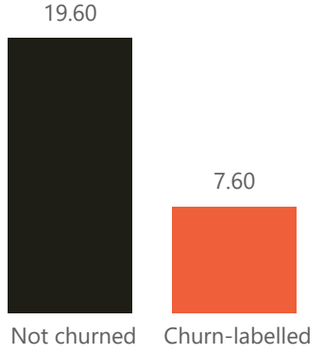


Customer Behaviour

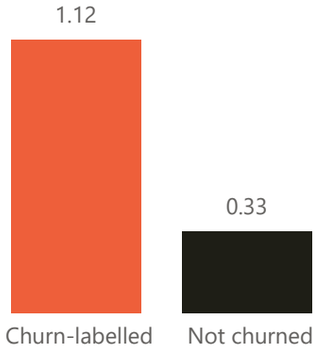
Customer Overview | Customer Behavior | Statistical Evidence | Insights & Recommendations

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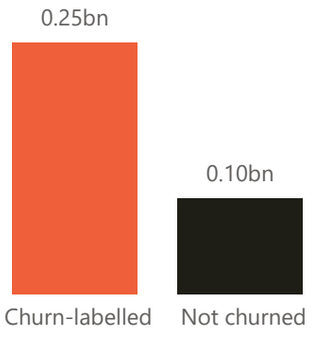
Average Months Since Last Transaction by Classification.



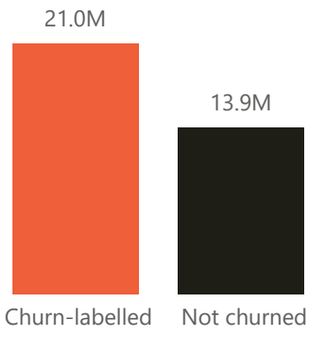
Average Active Cards by Customer Classification.



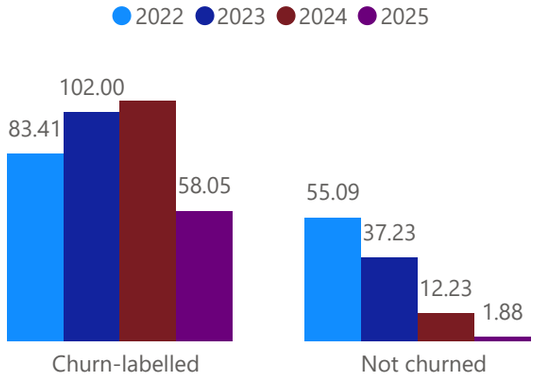
Average Total Billing Amount by Classification



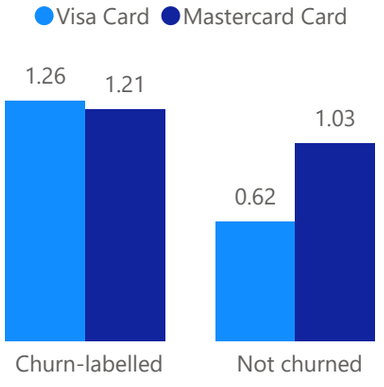
Average Customer Credit Limit by Classification



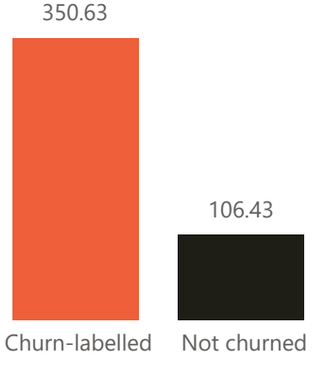
Average Annual Transactions by Customer Classification.



Average Visa and Mastercard Cards by Classification

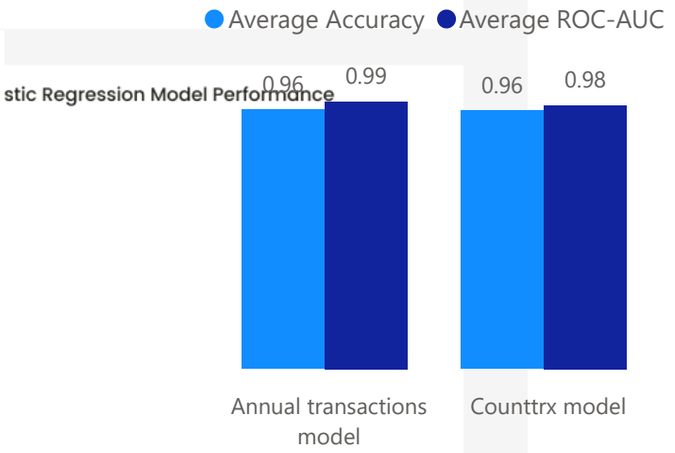


Average Transaction Count by Customer Classification



Statistical Evidence

Model	Accuracy	ROC-AUC	Not-Churned F1	Churn-Labelled F1
Annual transactions model	0.96	0.99	0.93	0.97
Counttrx model	0.96	0.98	0.92	0.97



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The annual transaction model achieved higher test accuracy and ROC-AUC than the Counttrx model. Both models showed strong classification performance on the test dataset. However, these models classify the dataset's supplied labels. Because the dataset does not document when churn was determined or the observation period, we cannot claim that they predict future customer churn.

Behavioural Statistical Tests

Variable	U Statistic	P-value	Rank-biserial correlation
Credit limit	9,481,181.00	2.5281E-106	0.26
Months since last transaction	21,585,032.00	<0.000001	-0.76
Total billing amount	6,088,074.00	<0.000001	0.53
Transaction count	1,235,735.50	<0.000001	0.90
Usage percentage	6,695,828.00	<0.000001	0.48

Demographic Statistical Tests

Variable	Chi-square	P-value	Cramér's V
Age band	9.40	0.02	0.03
Gender	0.20	0.65	0.00
Marital status	0.32	0.57	0.01

InsightS & Recommendations

Finding 1 — Customer Classification

72.05% of customers are classified as churned in the supplied dataset, compared with 27.95% classified as not churned.

This indicates a substantial imbalance in the supplied customer classifications.

Finding 2 — Transaction Behaviour

Customers labelled as churned recorded a higher average transaction count (350.63) than customers labelled as not churned (106.43).

The difference was statistically significant, with a very strong effect size.

Finding 3 — Card Ownership

Customers labelled as churned held an average of 1.12 active cards, compared with 0.33 among customers labelled as not churned.

The association between active card ownership and classification was strong.

Recommended Business Actions

1. Investigate transaction patterns
Review the transaction behaviour of both customer classifications to understand why the labelled groups show substantial differences.

2. Review card ownership
Investigate whether card ownership patterns reflect differences in customer engagement, product access or account usage.

3. Validate customer classification
Review the operational definition and timing of the churn labels before using the findings to design retention campaigns.

4. Improve customer data quality
Reconcile inconsistent card counts and transaction measures, and document missing values and variable definitions.

5. Strengthen future analysis
Collect time-stamped customer activity and verified churn outcomes to support genuine churn prediction and customer retention monitoring.

Dataset Limitations.

1. Undefined churn classification
The dataset does not provide a clear operational definition of churn or explain how customer classifications were assigned.

2. Unknown observation period
The data extraction date and observation window are not documented. Therefore, the timing of customer activity and churn cannot be established.

3. Incomplete date information
Some date fields contain only day and month, without the year, limiting reliable time-based analysis.

4. Missing values
Customer gender, marital status and last transaction information contain missing observations.

5. Model interpretation
The logistic regression models classify supplied customer labels. Their performance does not establish their ability to predict future churn.

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